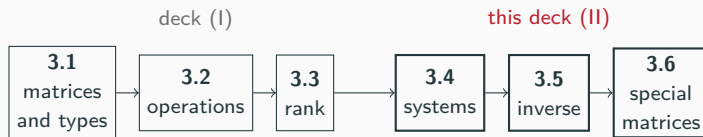


Linear Algebra

3 Matrices (II)

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Chapter map



Central idea: the rank decides whether a system has a solution; the inverse solves it in one go.

Exercises: Sheet 3, ex. 48–65.

3.4 Systems of linear equations

Matrix form of a system

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m \end{cases} \iff A\mathbf{x} = \mathbf{b},$$

with $A = [a_{ij}]$ the **coefficient matrix**, $\mathbf{x}$ the column of unknowns and $\mathbf{b}$ the column of constant terms. **Augmented matrix**: $[A | \mathbf{b}]$.

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with $A = [a_{ij}]$ the **coefficient matrix**, $\mathbf{x}$ the column of unknowns and $\mathbf{b}$ the column of constant terms. **Augmented matrix**: $[A | \mathbf{b}]$.

- **Inconsistent**: no solution. **Consistent**: at least one solution,
 - **with a unique solution**: exactly one; **with infinitely many solutions**: more than one.
- **Homogeneous**: $\mathbf{b} = \mathbf{0}$ (always consistent: it has the zero solution).

3.2: the system is consistent iff $\mathbf{b}$ is a linear combination of the columns of A .

Theorem

Let $A\mathbf{x} = \mathbf{b}$ have n unknowns, $r = \text{car } A$ and $s = \text{car}[A \mid \mathbf{b}]$.

- $r < s$: the system is **inconsistent**;
- $r = s = n$: **consistent with a unique solution**;
- $r = s < n$: **consistent with infinitely many solutions**, with $g = n - r$ free variables.

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Gaussian elimination:

1. row-reduce $[A \mid \mathbf{b}]$ by elementary operations *on rows*;
2. read r and s from the echelon form and classify;
3. if it is consistent, solve from the bottom up (back substitution).

Example: a system with a unique solution

$$C\mathbf{x} = (2, -11, -9):$$

$$\left[\begin{array}{ccc|c} 2 & 0 & 0 & 2 \\ -6 & 2 & -3 & -11 \\ -6 & 0 & -1 & -9 \end{array} \right] \xrightarrow[L_3 \rightarrow L_3 + 3L_1]{L_2 \rightarrow L_2 + 3L_1} \left[\begin{array}{ccc|c} 2 & 0 & 0 & 2 \\ 0 & 2 & -3 & -5 \\ 0 & 0 & -1 & -3 \end{array} \right]$$

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$r = s = 3 = n$: consistent with a unique solution. From the bottom up:

$$-x_3 = -3 \Rightarrow x_3 = 3, \quad 2x_2 - 9 = -5 \Rightarrow x_2 = 2, \quad 2x_1 = 2 \Rightarrow x_1 = 1.$$

$\mathbf{x} = (1, 2, 3)$ (agrees with 3.2: $C(1, 2, 3) = (2, -11, -9)$).

Example: analysis with a parameter

$$\begin{cases} x + y = 1 \\ 2x + ky = 3 \end{cases} : \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & k & 3 \end{array} \right] \xrightarrow{L_2 \rightarrow L_2 - 2L_1} \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & k-2 & 1 \end{array} \right].$$

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- $k \neq 2$: $r = s = 2 = n \Rightarrow$ **unique solution**: $y = \frac{1}{k-2}$, $x = \frac{k-3}{k-2}$.
- $k = 2$: the last row is $[0 \ 0 \mid 1]$, i.e. $0 = 1$: $r = 1 < s = 2 \Rightarrow$ **inconsistent**.

With parameters, the special cases are the values that make a pivot zero.

Theorem

If $\mathbf{x}_p$ is a solution of $A\mathbf{x} = \mathbf{b}$, then *all* the solutions are

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h, \quad \mathbf{x}_h \text{ a solution of the homogeneous system } A\mathbf{x} = \mathbf{0}.$$

Why: if $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{x}_p = \mathbf{b}$, then $A(\mathbf{x} - \mathbf{x}_p) = \mathbf{0}$. The solutions of the homogeneous system form a subspace (Ch. 2).

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Example. $\begin{cases} x + y - z = 1 \\ x - y + z = 1 \end{cases}$: one solution is $(1, 0, 0)$; the homogeneous system gives $x = 0$,
 $y = z$.

$$\mathbf{x} = (1, 0, 0) + t(0, 1, 1), \quad t \in \mathbb{R}.$$

- **Ex. 48** solving by Gaussian elimination.
- **Ex. 49–51** analysis with parameters. Look for the values that make pivots zero.
- **Ex. 53–54** general solution = one known solution + solutions of the homogeneous system.

3.5 Inverse

Definition

A square A of order n is **invertible** if there exists A^{-1} such that $AA^{-1} = A^{-1}A = I_n$. If it exists, it is unique.

Theorem

A is invertible $\iff \text{car } A = n$ (A non-singular) $\iff$ the columns of A are lin. indep.

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Properties (A, B invertible of the same order):

$$(A^{-1})^{-1} = A, \quad (AB)^{-1} = B^{-1}A^{-1}, \quad (A^T)^{-1} = (A^{-1})^T, \quad (\alpha A)^{-1} = \frac{1}{\alpha}A^{-1} \ (\alpha \neq 0).$$

If A is invertible and $AB = 0$, then $B = A^{-1}AB = 0$.

Computing the inverse (Gauss-Jordan)

Method: $[A \mid I] \xrightarrow{\text{row operations}} [I \mid A^{-1}]$. If a zero row appears on the A side, A is not invertible.

Computing the inverse (Gauss-Jordan)

Method: $[A | I] \xrightarrow{\text{row operations}} [I | A^{-1}]$. If a zero row appears on the A side, A is not invertible.

$$\left[\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & 0 & 0 \\ -6 & 2 & -3 & 0 & 1 & 0 \\ -6 & 0 & -1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & -3 & 3 & 1 & 0 \\ 0 & 0 & -1 & 3 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 & -3 & \frac{1}{2} & -\frac{3}{2} \\ 0 & 0 & 1 & -3 & 0 & -1 \end{array} \right]$$

In the last step: $L_3 \rightarrow -L_3$; $L_2 \rightarrow (L_2 + 3L_3)/2$; $L_1 \rightarrow L_1/2$.

$$C^{-1} = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ -3 & \frac{1}{2} & -\frac{3}{2} \\ -3 & 0 & -1 \end{bmatrix} \quad (\text{check: } CC^{-1} = I_3).$$

If A is invertible:

$$A\mathbf{x} = \mathbf{b} \iff \mathbf{x} = A^{-1}\mathbf{b}.$$

Example. $C\mathbf{x} = (2, -11, -9)$: $\mathbf{x} = C^{-1}(2, -11, -9) = (1, 2, 3)$, as in 3.4.

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Matrix equations: order matters

$AX = B \Rightarrow X = A^{-1}B$ (multiply *on the left*); $XA = B \Rightarrow X = BA^{-1}$ (on the *right*).

Never 'divide' by a matrix; and $A^{-1}B \neq BA^{-1}$ in general.

Ch. 2, change of basis: if the columns of P are the vectors of the basis B , then $[\mathbf{v}]_B = P^{-1}\mathbf{v}$.

- **Ex. 52** inverses by Gauss-Jordan. In c), compare the inverse with the transpose.
- **Ex. 60–61** systems via the inverse.
- **Ex. 62** a matrix equation: isolate X , minding the order.
- **Ex. 63–64** inverse, systems and negative powers.

3.6 Special matrices

Symmetric and skew-symmetric

Definitions (A square)

Symmetric: $A^T = A$ ($a_{ji} = a_{ij}$). **Skew-symmetric:** $A^T = -A$ ($a_{ji} = -a_{ij}$; zero diagonal).

$$\begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} \text{ symmetric,} \quad \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix} \text{ skew-symmetric.}$$

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Every square matrix is the sum of a symmetric and a skew-symmetric matrix:

$$A = \underbrace{\frac{1}{2}(A + A^T)}_{\text{symmetric}} + \underbrace{\frac{1}{2}(A - A^T)}_{\text{skew-symmetric}}, \quad \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Complex matrices: Hermitian

For A with complex entries, the **conjugate transpose** is $A^* = \overline{A}^T$ (conjugate every entry and transpose).

Definitions

Hermitian: $A^* = A$. **Skew-Hermitian:** $A^* = -A$.

$$\begin{bmatrix} 2 & 1-i \\ 1+i & 3 \end{bmatrix} \text{ Hermitian (real diagonal), } \quad \begin{bmatrix} i & 2 \\ -2 & 0 \end{bmatrix} \text{ skew-Hermitian (purely imag. diagonal).}$$

A real matrix is Hermitian iff it is symmetric. As before: $A = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*)$.

Orthogonal matrices

Definition

A real square A is **orthogonal** if $A^T A = A A^T = I$, i.e. $A^{-1} = A^T$.

Example. The rotations R_θ (3.2) are orthogonal:

$$R_\theta^T R_\theta = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = I_2, \quad R_\theta^{-1} = R_{-\theta} = R_\theta^T.$$

Undoing a rotation means rotating the opposite way — and the inverse comes with no computation, just by transposing.

- **Ex. 55** classifying matrices. Some are none of the four types.
- **Ex. 56–59** properties of the transpose and the conjugate transpose; decompositions.
- **Ex. 65** orthogonal matrices.

LINEAR ALGEBRA ON THE COMPUTER: rank, inverse, systems

For illustration only — not examinable. Python (NumPy); in MATLAB/Octave: `rank(C)`, `inv(C)`, `C\b`.

```
import numpy as np
C = np.array([[ 2, 0,  0],
              [-6, 2, -3],
              [-6, 0, -1]])
np.linalg.matrix_rank(C)    # 3
np.linalg.inv(C)
# [[ 0.5  0.   0. ]
#  [-3.   0.5 -1.5]
#  [-3.   0.  -1. ]]
np.linalg.solve(C, [2,-11,-9])
# [1.  2.  3.]
```

The same results we obtained by hand with the matrix C : $\text{car } C = 3$ (3.3), C^{-1} by Gauss–Jordan (3.5) and $\mathbf{x} = (1, 2, 3)$ (3.4). `solve` uses Gaussian elimination, **without computing the inverse**: in general, this is more efficient and numerically more stable. Computing A^{-1} just to solve $A\mathbf{x} = \mathbf{b}$ is wasteful.

Results are in floating point: $-0.$ or errors of the order of 10^{-16} may appear.

Quick reference — Chapter 3

Concept	How to obtain it / what it says
Product AB	row by column; $(m \times n)(n \times p) = m \times p$; $AB \neq BA$ in general
$A\mathbf{x}$	linear combination of the columns of A
Transpose	$(AB)^T = B^T A^T$
Rank	max. no. of lin. indep. rows (= columns); non-zero rows of the echelon form
Elementary operations	swap; multiply by $\alpha \neq 0$; add a multiple of another row
System $A\mathbf{x} = \mathbf{b}$	$r < s$ inconsistent; $r = s = n$ unique solution; $r = s < n$ infinitely many ($g = n - r$)
Solutions	$\mathbf{x} = \mathbf{x}_p +$ (solutions of the homogeneous system)
Inverse	exists iff $\text{car } A = n$; $[A I] \rightarrow [I A^{-1}]$; $(AB)^{-1} = B^{-1}A^{-1}$
Matrix equations	$AX = B \Rightarrow X = A^{-1}B$; $XA = B \Rightarrow X = BA^{-1}$
Symmetric / skew-sym.	$A^T = A$ / $A^T = -A$
Hermitian / skew-Herm.	$A^* = A$ / $A^* = -A$, with $A^* = \overline{A}^T$
Orthogonal	$A^T A = I$, i.e. $A^{-1} = A^T$

KEY TAKEAWAYS — Chapter 3

1. A matrix arranges vectors in columns; multiplying it by a vector means forming a linear combination of those columns.
2. The matrix product composes operations: it depends on the order and exists only for compatible sizes.
3. The rank measures how many rows (or columns) are genuinely independent, and is computed by bringing the matrix to echelon form.
4. Comparing the rank of the system's matrix with that of the augmented matrix tells you at once whether the system is inconsistent, has a unique solution or has infinitely many.
5. A square matrix is invertible exactly when it has full rank; in that case, the system is solved by multiplying by the inverse.

Next (Ch. 4): determinants — a single number that decides whether a matrix is invertible.

References (full list: course bibliography)

- Santana, A. P., Queiró, J. F., *Introdução à Álgebra Linear*, Gradiva (in Portuguese).
- Lipschutz, S., Lipson, M., *Schaum's Outline of Linear Algebra*, McGraw-Hill — many solved exercises.
- Strang, G., *Introduction to Linear Algebra*, Wellesley-Cambridge Press.

Resources — Sheet 3 (exercises 39–65) and its solutions.